1 Fig. 1 shows the probability distribution of the discrete random variable X.

x	1	2	3	4	5
$\mathrm{P}(X = x)$	0.2	0.1	k	$2k$	$4k$

Fig. 1

(a) Find the value of k. **[2]**

(b) Find $\mathrm{P}(X \neq 4)$. **[2]**